



ELEMENTARY MATH (CALCULUS)



DEFINITE INTEGRALS: VOLUME OF SOLIDS OF REVOLUTION



LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- identify problems associated with volumes of objects by means of integration;
- obtain the volumes of objects by integration, using disc method;
- obtain the volumes of objects by integration, using washer method; and
- obtain the volumes of objects by integration, using shell method.





INTRODUCTION

In this note, the applications of definite integrals to calculating volumes of solid shapes are considered. The idea will be to dissect the three dimensional objects into pieces that resemble discs or shells, whose volumes can be approximated. The volume of the object is obtained in the form of disks or a set of embedded shells, integrated over given intervals.





VOLUMES BY INTEGRATION

If a region in the plane is revolved about a given line, the resulting solid is a solid of revolution, and the line is called the axis of revolution. When calculating the volume of a solid generated by revolving a region bounded by a given function about an axis, follow the steps below:

1. Sketch the area and determine the axis of revolution, (this determines the variable of integration).
2. Sketch the cross-section, (disc, shell, washer) and determine the appropriate formula.
3. Determine the boundaries of the solid.
4. Set up the definite integral, and integrate.





VOLUMES BY INTEGRATION: DISC METHOD

The volume of a solid of revolution is given by

$$V = \pi \int_a^b [R(x)]^2 dx$$

where $R(x)$ is the region bounded by $y = f(x)$ revolved the x -axis, and

$$V = \pi \int_a^b [R(y)]^2 dy$$

where $R(y)$ is the region bounded by $x = f(y)$ revolved the y -axis.





VOLUMES BY INTEGRATION: DISC METHOD

Example 1.

Let R be the region bounded by the curve $y = f(x) = (-x + 1)$, the x -axis, and the lines $x = 0$ and $x = 1$. Find the volume of the solid of revolution obtained by revolving R about the x -axis.

Solution

Using the formula

$$V = \pi \int_a^b [R(x)]^2 dx$$

where $R(x) = (-x + 1)$ over $a = 0$, and $b = 1$, we have

$$\begin{aligned} V &= \pi \int_0^1 [(-x + 1)]^2 dx = \pi \left[\frac{-(-x + 1)^3}{3} \right]_0^1 \\ &= \pi \left(\frac{-(-1 + 1)^3}{3} - \left(\frac{-(-0 + 1)^3}{3} \right) \right) = \frac{\pi}{3} \end{aligned}$$

Therefore, the volume is $\frac{\pi}{3}$ cubic units.





VOLUMES BY INTEGRATION: DISC METHOD

Example 2.

Find the volume of the solid of revolution obtained by revolving the region between the parabola $x = y^2 + 1$, and the lines $x = 3$ about the line.

Solution

Setting $y^2 + 1 = 3$, we have $x = R(y) = 2 - y^2$, so that $a = -\sqrt{2}$ and $b = \sqrt{2}$. Therefore,

$$\begin{aligned} V &= \pi \int_{-\sqrt{2}}^{\sqrt{2}} [(2 - y^2)]^2 dy = \pi \int_{-\sqrt{2}}^{\sqrt{2}} [4 - 4y^2 + y^4]^2 dy \\ &= \pi \left[4y - \frac{4y^3}{3} + \frac{y^5}{5} \right]_{-\sqrt{2}}^{\sqrt{2}} = \frac{64\sqrt{2}\pi}{15} \end{aligned}$$

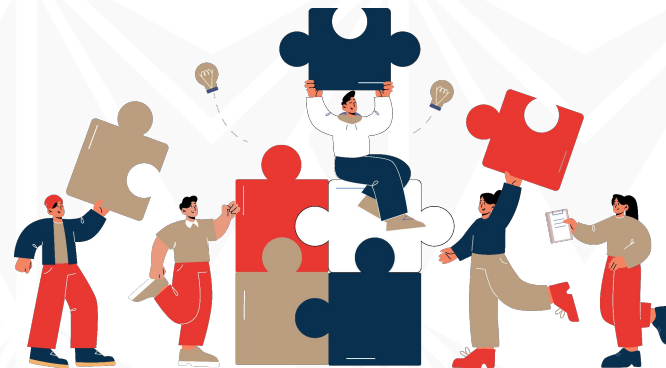
Hence, the volume is $\frac{64\sqrt{2}\pi}{15}$ cubic units.





TASK

Find the volume of the solid of revolution obtained by revolving $f(x) = \sqrt{x}$, the x -axis, and the lines $x = 1$ and $x = 4$ about the x -axis.





VOLUMES BY INTEGRATION: WASHER METHOD

If a solid of revolution has any hollow spots in it, we use the washer method. It is of the form of a circular washer of outer radius R and inner radius r . In this case, we have

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx$$

and

$$V = \pi \int_a^b ([R(y)]^2 - [r(y)]^2) dy,$$

accordingly.





VOLUMES BY INTEGRATION: DISC METHOD

Example 1.

Find the volume of the solid formed by revolving the region bounded by the graphs of $y = x^2$, $y = x^5$, $x = 0$, and $x = 1$ about the x -axis.

Solution

Here, $R(x) = x^2$, and $r(x) = x^5$, so that

$$\begin{aligned} V &= \pi \int_0^1 ([x^2]^2 - [x^5]^2) dx = \pi \int_0^1 (x^4 - x^{10}) dx \\ &= \pi \left[\frac{x^5}{5} - \frac{x^{11}}{11} \right]_0^1 = \frac{6\pi}{55} \end{aligned}$$

Hence, the volume is $\frac{6\pi}{55}$ cubic units.





VOLUMES BY INTEGRATION: WASHER METHOD

Example 2.

Find the volume of the solid formed by revolving the region bounded by the graphs of $x = y^2$, $x = 3$, and $x = 0$ about the y -axis.

Solution

Here, $R(y) = 3$, and $r(y) = y^2$, so that

$$\begin{aligned} V &= \pi \int_0^{\sqrt{3}} (3^2 - [y^2]^2) dy = \pi \int_0^{\sqrt{3}} (9 - y^4) dy \\ &= \pi \left[9y - \frac{y^5}{5} \right]_0^{\sqrt{3}} = \frac{36\sqrt{3}\pi}{5} \end{aligned}$$

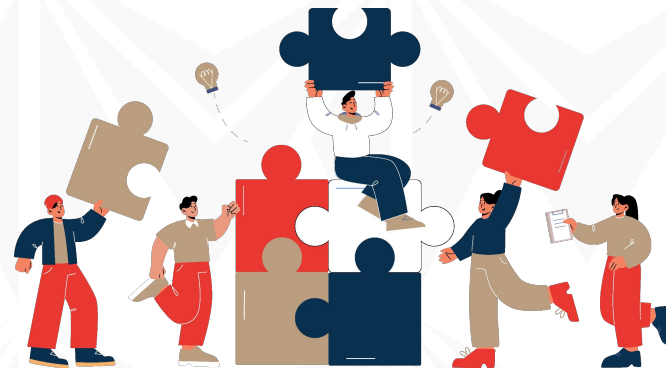
Hence, the volume is $\frac{36\sqrt{3}\pi}{5}$ cubic units.





TASK

Find the volume of the solid formed when the region under the curve $y = x^2$ over the interval $[0, 2]$ is rotated about the line $y = -1$.





VOLUMES BY INTEGRATION: SHELL METHOD

The shell method is used when integrating along an axis parallel to the axis of revolution. In this case, the volume is given by

$$V = 2\pi \int_a^b xf(x)dx$$

where x is the radius of revolution, and

$$V = 2\pi \int_a^b yf(y)dy$$

where y is the radius of revolution.





VOLUMES BY INTEGRATION: SHELL METHOD

Example 1.

Find the volume of the solid obtained by rotating about the y -axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$.

Solution

Given that $f(x) = 2x^2 - x^3$, then setting $f(x) = 0$, we have $x = 0$ and $x = 2$. Therefore, the volume is

$$\begin{aligned} V &= 2\pi \int_0^2 x(2x^2 - x^3)dx = 2\pi \int_0^2 (2x^3 - x^4)dx \\ &= 2\pi \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 = 2\pi \left(8 - \frac{32}{5} \right) = \frac{16\pi}{5} \end{aligned}$$

Hence, the volume is $\frac{16\pi}{5}$ cubic units.





VOLUMES BY INTEGRATION: SHELL METHOD

Example 2.

Use the shell method to find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \sqrt{x}$ from 0 to 1.

Solution

Given that $y = \sqrt{x}$, we have $x = y^2$ so that $y = 0$, $y = 1$ and $f(y) = 1 - y^2$.

Therefore, the volume is

$$\begin{aligned} V &= 2\pi \int_0^1 y(1 - y^2)dy = 2\pi \int_0^1 (y - y^3)dy = 2\pi \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1 \\ &= 2\pi \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{\pi}{2} \end{aligned}$$

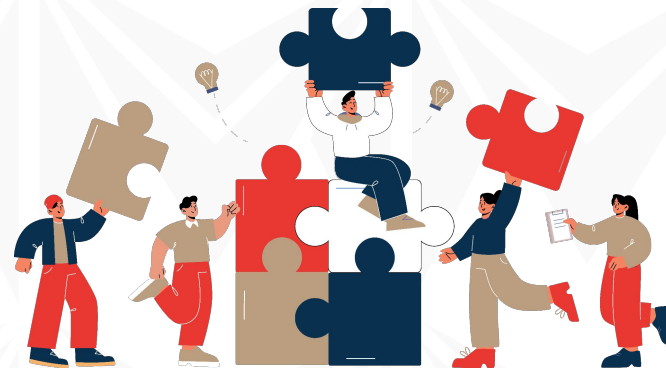
Hence, the volume is $\frac{\pi}{2}$ cubic units.





TASK

Find the volume of the solid obtained by rotating about the y -axis the region between $y = x$ and $y = x^2$.





SUMMARY

The applications of definite integrals to calculating volumes of solid shapes are considered.

The volume of a solid is obtained by integration using the disc, washer, and shell methods.

Using the disc method, the volume is given by

$$V = \pi \int_a^b [R(x)]^2 dx.$$

Using the washer method, the volume is given by

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx.$$

Using the shell method, the volume is given by

$$V = 2\pi \int_a^b xf(x) dx.$$





FURTHER READING:

"Calculus Know-It-ALL " by Stan Gibilisco.

"Theory and Problems of ADVANCED CALCULUS" by Robert Wrede,
and Murray R. Spiegel.

"Advanced Engineering Mathematics" by H. K. Dass.

"Engineering Mathematics" by K. A. Stroud.





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THANK YOU